

Real Estate Tokenization: Exploring Definitions, Mechanisms, and Future Applications

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Abstract— The growing use of blockchain technology has introduced new opportunities in real estate investment by addressing key challenges such as high capital requirements, limited transparency, and low liquidity. Real estate tokenization, which divides property assets into digital tokens on a blockchain, has become a viable solution. While previous studies have focused on technical design, legal issues, or financial aspects, few have explored how tokenization is implemented in practice or why Ethereum token standards such as ERC-20, ERC-721, and ERC-1400 remain dominant despite newer options. This study investigates common tokenization models, platform-specific practices, and the rationale behind token standard choices through a comprehensive review of recent literature. Findings reveal that widely adopted standards are preferred due to their simplicity, regulatory compatibility, and strong developer support. In contrast, more advanced standards like ERC-777 and ERC-1155 offer advanced features, their adoption remains limited due to complexity and ecosystem immaturity. These insights highlight why established standards prevail today, while newer standards show strong potential for future real estate models focused on automation and flexibility.

Keywords— Real Estate Tokenization, Blockchain, Smart Contracts

I. INTRODUCTION

In recent years, the advent of blockchain has emerged as one of the most revolutionary innovations. Nowadays, it is recognized as a versatile tool, gaining widespread attention in reshaping industries on its adoption across various sectors, from finance and supply chain management to healthcare and government services [9][15][21].

Despite advancements in other industries, most individuals find real estate investment difficult due to high entry costs, limited transparency, and low liquidity [19]. These barriers often prevent small investors from participating in the market. Additionally, rising property prices in major urban areas further reduce accessibility, especially for first-time or low-capital investors [08].

The concept of fractional ownership in real estate has existed historically through models like REC or REITs, though these often lack flexibility and liquidity [14][19] and complex regulatory frameworks [25][23]. However, these limitations opened the door to more innovative solutions.

With blockchain, real estate tokenization has become a transformative concept [5][22]. This innovation ‘democratizes’ access to real estate investments for individuals to invest in property ownership with minimal capital [13][20]. Early implementations between 2017 and 2019 saw various asset types, from ski resorts to Manhattan apartments, adopt this approach [12][14].

While this concept holds great promise, existing studies have primarily focused on technological challenges [24],

financial potential [6][16], and legal aspects [4][11] but little attention to understanding its applications.

This research aims to explore (1) the general & comprehensive mechanism of the concept of real estate tokenization, (2) a comparison of how existing platforms perform in practice, and (3) the reasons behind the prevalent use of ERC-20, ERC-721, and ERC-1400 in real estate tokenization, and why are alternative ERC standards not widely adopted.

A. Evolution, Key Concepts, and Terminologies of Real Estate Tokenization

The advent of blockchain technology has redefined how real estate assets are owned and traded. These tokens can represent fractional ownership, equity shares, or revenue distribution rights, allowing for broader market participation and more efficient transactions [24]. Real estate tokenization has gained traction through platforms such as RealT [10], BrickMark [22], and BlockEstate [7], demonstrating its real-world application.

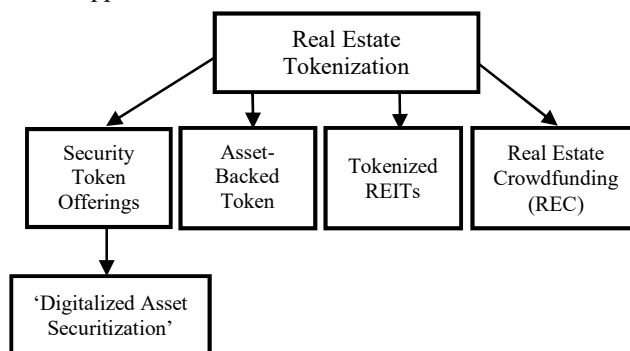


Fig. 1. Terms related to real estate tokenization.

FINMA initially referred to real estate tokenization as Digital Asset Securitization, which later evolved into Security Token Offerings (STOs) and Asset-Backed Tokens (ABTs), representing digital tokens linked to physical assets like real estate [14][19][25][23]. More recently, Tokenized REITs (T-REITs) have emerged, blending REIT structures with blockchain-based tokens [12].

Before blockchain, Real Estate Crowdfunding (REC) introduced fractional ownership via centralized platforms. Although it improved accessibility, REC lacked liquidity and remained intermediary-dependent [12][14].

II. RESEARCH APPROACH

This study aims to empirically analyze the implementation and practical implications of real estate tokenization. To achieve this, a systematic literature review (SLR) is conducted alongside supplementary analysis to bridge theoretical concepts with existing platforms, guided by the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-

Analysis) framework to ensure the articles' transparency and reliability [17][18].

Existing definitions regarding real estate tokenization are systematically reviewed to answer the first research question using academic databases such as IEEE Xplore, Springer Link, ScienceDirect, Emerald, and ACM, alongside recognized preprint servers (arXiv), university repositories (sbs.ox.ac.uk, and others), and indexed conference proceedings (CEUR-WS). Papers were selected based on relevance to the research topic rather than restricted to specific publishers with corresponding keywords like "tokenization", "real estate tokenization", "blockchain", "blockchain fractional ownership", "smart contract", "ERC token standards", that are published between 2020 and 2025.

The initial search yielded around 176 articles using Publish or Perish. For a more structured and evidence-based screening process, the PRISMA framework was used and shown in "Fig. 2". The first screening process manually filtered the articles by title and abstract for alignment with the research objectives, primarily focusing on studies of definitions, mechanisms, and potential challenges and benefits of real estate tokenization which then will be evaluated for its practical applications. The second-level screening reviewed full texts of eligible articles based on inclusion and exclusion criteria, as shown in "Table 1". Peer-reviewed studies on blockchain applications in real estate and those addressing technological, financial, or practical aspects of tokenization were retained.

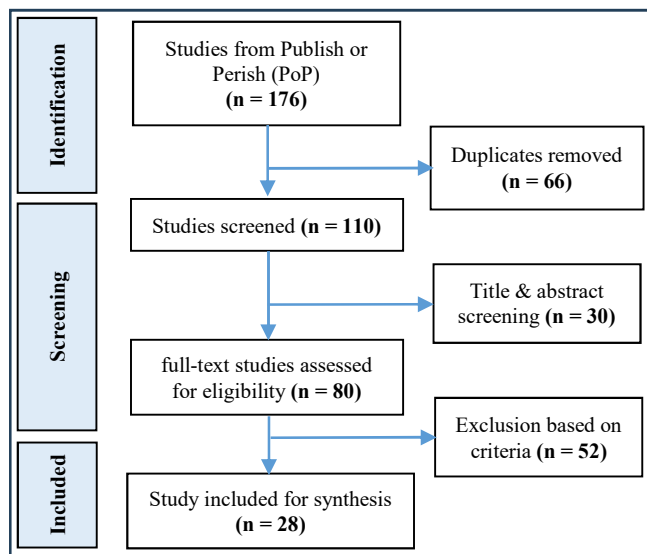


Fig. 2. PRISMA flowchart for article screening.

Meanwhile, irrelevant topics and purely theoretical studies were excluded from non-peer-reviewed sources. This laborious screening and filtering process resulted in 28 selected articles with over 300 pages.

This analysis explores how real estate tokenization works by converting property into fractional digital tokens via blockchain. It examines the process, including legal structuring (e.g., SPVs/LLCs), smart contract automation, and the use of Ethereum standards like ERC-20, ERC-721, and ERC-1400 for ownership, compliance, and tradability. It then compares these standards and evaluates their suitability and future potential in real estate tokenization.

TABLE I. INCLUSION AND EXCLUSION CRITERIA

No	Inclusion Criteria	Exclusion Criteria
1	Peer-reviewed articles from 2020-2025	Non-peer-reviewed content like promotional purposes or unrelated blog
2	Studies on blockchain, smart contracts, and tokenization in real estate	Studies not focused on real estate, tokenization, or any mentioned keyword(s)
3	Articles can be publicly accessed via a common academic indexer	Articles require privileged user access
4	Articles written in English	Duplicate studies or papers with minimal contributions

The findings from the selected paper are then compared with the general mechanism derived from various academic references to assess alignment between theory and implementation, which later will be aligned and concluded with the initial hopes and dreams of investors as part of this innovative concept.

III. MECHANISMS OF REAL ESTATE TOKENIZATION

A. End-to-End Process of Real Estate Tokenization

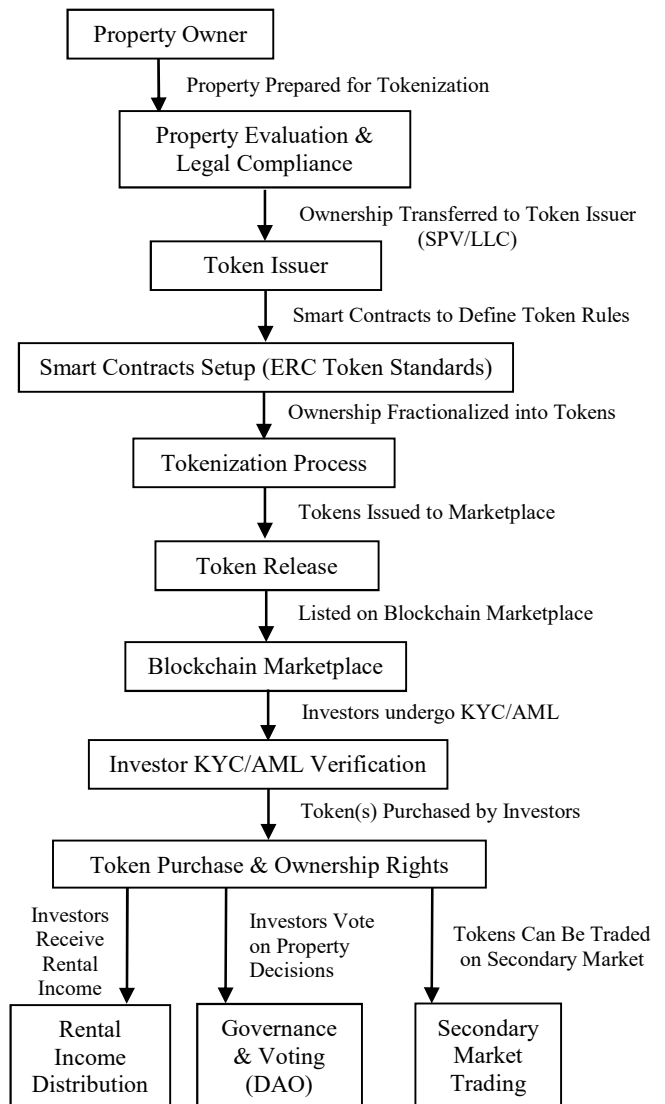


Fig. 3. Real estate tokenization mechanism flow.

The process begins with property evaluation and legal compliance to ensure the asset is suitable for fractional ownership [2]. The ownership is then transferred to a Special Purpose Vehicle (SPV), a legal entity created specifically to hold the property and issue tokens, which acts as the token issuer, keeping the original property certificate and creating digital tokens representing fractional ownership [14].

A smart contract based on Ethereum's token standards is deployed to facilitate this, as shown in "Table 2" which defines the token rules, ownership rights, and compliance mechanisms [2].

ERC, short for Ethereum Request for Comment, outlines specifications for how tokens function on the Ethereum network [27]. Each token standard has its functionality and characteristics. ERC-20, ERC-721, and ERC-1400 are commonly used for real estate tokenization cases.

1) Token Standard

TABLE II. TOKEN STANDARDS IN ETH NETWORK

Token Type	Token Standards	Description	Ref
Fungible Token (FT)	ERC-20, ERC-1155, ERC-1400, ERC-777, ERC-223,	Interchangeable tokens adapt to asset features and capabilities for crypto marketplaces, depending on the standard.	[1][2][16][22][25][26][27][28]
Non-Fungible Token (NFT)	ERC-721, ERC-5505	NFTs in marketplaces hold data on underlying assets, which aren't shareable among investors but reflect collective ownership.	[1][16][27][28]
Governance Token (GT)	ERC-20, ERC-2767	Interchangeable tokens are for use in governance and decision-making structures and DAOs.	[16]

The total token supply is determined by dividing the property's value by the price per token. For example, a property worth IDR 2 billion, if each token is valued at IDR 1 million, would result in 2,000 tokens available for sale, simply as the formula shown below:

$$\text{Total Tokens} = \frac{\text{Property Value}}{\text{Token Unit Value}}$$

Fig. 4. Total token(s) equations formula.

Once the tokens are issued, they are listed on a marketplace platform where investors undergo KYC/AML verification before purchasing [3][14]. Investors gain fractional ownership rights to receive rental income distributions, participate in governance decisions, and trade tokens on secondary markets [2].

Secondary Market Trading refers to the ability of investors to trade their real estate tokens on a blockchain marketplace after the initial issuance. This allows investors to buy and sell tokens without waiting for the property itself to be sold [2]. In traditional real estate, liquidity is a significant issue, but tokenization enables fractional ownership and seamless transactions, much like stock exchanges [14].

B. Comparative Analysis of Real Estate Tokenization Mechanisms in Existing Platforms

Existing real estate tokenization platforms vary from what is shown in "Fig 3" regarding legal structuring, blockchain standards, and secondary market accessibility. RealT utilizes a Series LLC structure to hold tokenized properties for regulatory compliance [10], whereas BrickMark employs a Special Purpose Vehicle (SPV) for asset governance and token issuance [23].

Blockchain implementation also differs. BrickMark uses the ERC-20 standard on Ethereum to issue tokenized perpetual bonds, while RealT likely relies on ERC-20 tokens as well, given their compatibility with decentralized exchanges like Uniswap [10][23]. BlockEstate, by contrast, uses a private blockchain model based on Hyperledger, focusing on institutional compliance and omitting public token trading. It manages ownership transfers via chaincode and implements a private ledger architecture to enhance transactional integrity [7].

Secondary market integration varies significantly. RealT enables open trading on DEXs, BrickMark uses controlled private marketplaces, and BlockEstate excludes decentralized markets in favor of institutional frameworks [7][10][23].

TABLE III. ERC TOKEN STANDARDS CHARACTERISTICS

ERC Standard	Main Usage	Benefits	Limitations	Ref
ERC-20	Fractional ownership of properties, liquidity, tradability	Simple, highly interoperable, widely adopted	No support for unique assets, no regulatory features	[10][22][25]
ERC-721	Tokenizing unique properties or assets	Provides uniqueness & traceability, ideal for luxury properties	Not fungible, less liquidity compared to ERC-20	[7][10][25][27]
ERC-1400	Security tokens with regulatory compliance (KYC/AML)	Built-in compliance features, suitable for regulated markets	More complex to implement, not as widely adopted	[7][25]
ERC-777	Advanced fungible tokens with better functionality	Support more advanced token interactions (e.g., hooks, more efficient than ERC-20)	Complex; adoption is lower	[2][16]
ERC-1155	Mixed-use cases (both fungible & non-fungible tokens)	More efficient for mixed asset types,	Complexity in implementation	[22]
ERC-223	Safer token transfers	Prevent "lost tokens" due to failed transfers	Less adoption, similar to ERC-20	[26]

1) Functional and Regulatory Alignment

"Table 3" shows that while several ERC standards offer advanced features, ERC-20, ERC-721, and ERC-1400 are preferred due to their compatibility with three core pillars of real estate tokenization: fractional ownership, asset uniqueness, and regulatory compliance.

ERC-20 remains the most widely adopted standard, due to its fungible nature that facilitates the division of property

value into equal units. This characteristic makes it well-suited for fractional ownership, as well as simplifies investor onboarding, and enhances liquidity through broad compatibility with digital wallets and decentralized exchanges [2][10][22][25].

ERC-721, by contrast, is non-fungible and designed to represent unique, indivisible assets. Its structure makes it well-suited for tokenizing entire properties or distinctive real estate units where traceability and uniqueness are critical. However, the lack of native support for fractionalization limits its applicability to high-value assets or whole-property transactions [1][25][27][28].

ERC-1400 on the other hand, is a security token that expands upon the ERC-20 standard while adding functionality for essential compliance and regulatory needs. These compliance features are necessary for real estate markets where KYC/AML obligations and investor qualification rules must be enforced natively on-chain [25].

IV. FUTURE PERSPECTIVES AND INNOVATIONS

Building upon the gaps identified in ERC-20, ERC-721, and ERC-1400, this thesis proposes that newer Ethereum standards, specifically ERC-777 and ERC-1155, can serve as forward-looking solutions for real estate tokenization. The contribution of this section is to position this standards not only as technically advanced protocols, but as mechanisms that directly address compliance, automation, and multi-asset management challenges faced in current implementations.

While prior studies, such as Gupta (2020) and Joshi (2022), have demonstrated the feasibility of applying ERC-777 and ERC-1155 to real estate contexts, they did not explicitly link these standards to the shortcomings of earlier approaches. This thesis extends their insights by framing ERC-777 and ERC-1155 as structured solutions to the observed gaps, thereby offering a practical roadmap for future adoption.

A. ERC-777: Allowing Compliance-Aware Automation

This thesis proposes ERC-777 as a solution to compliance and automation gaps left by ERC-20 in real estate tokenization. Unlike ERC-20, ERC-777 enables advanced functions such as operator roles and send/receive hooks, which allow business logic to be embedded directly into token transactions. In the real estate context, this means compliance checks (e.g., KYC/AML verification), regulatory validation through oracles, and automated dividend distribution can be enforced at the token level. Such features make ERC-777 well-suited for tokenized property markets that require both fractional ownership and strict regulatory adherence [1][16][27].

Prior work by Gupta (2020) demonstrated ERC-777's general utility in real estate tokenization, and this thesis extends that insight by explicitly positioning ERC-777 as a targeted response to ERC-20's limitations in compliance automation and investor protection.

B. ERC-1155: Streamlining Multi-Asset Management

This thesis also proposes ERC-1155 as a forward-looking solution to the rigidity of ERC-20 and ERC-721 in handling diverse real estate assets. Unlike the one-contract-per-token model of ERC-20 and ERC-721, ERC-1155 allows both fungible and non-fungible tokens to coexist under a single

smart contract, reducing gas costs and simplifying portfolio management.

In real estate tokenization, this enables practical use cases such as tokenizing mixed-use properties (residential + commercial), combining fractional ownership shares with unique property deeds, and managing diversified REIT-style portfolios more efficiently [22][25][27].

Joshi and Choudhury (2022) highlighted ERC-1155's efficiency for real estate applications, but this thesis builds on their work by framing ERC-1155 as a structured response to the scalability and flexibility issues inherent in ERC-20 and ERC-721. While adoption in real estate remains limited, ERC-1155's potential for streamlining multi-asset management and reducing contract complexity makes it a strong contender for future use in real estate investment trusts (REITs) and tokenized asset portfolios [22].

C. Toward Hybrid and Smart Token Standards

According to A. Gupta and S. Joshi works, they discussed that both ERC-777 and ERC-1155 implemented the InterPlanetary File System (IPFS) protocol to ensure documentation immutability and transparency. Upon administrator approval, tokenized property records can be linked to IPFS for verification, enhancing auditability and trust.

Moreover, ERC-777 enhances ERC-20 by enabling operator-based delegation and transaction hooks, which support automated functions such as KYC/AML regulatory checks and dividend distribution. The way A. Gupta et al. (2020) implement the ERC-777 standard for real estate tokenization highlights its suitability for fractional ownership and dividend distribution via smart contracts.

Although the paper does not detail the internal mechanics of ERC-777, the standard itself supports advanced features such as operator management and transaction hooks, which offer greater flexibility and automation compared to ERC-20.

In addition, A. Gupta et al. also proposed a slightly different approach regarding the workflow of the overall tokenization compared to "Fig. 3" as shown below.

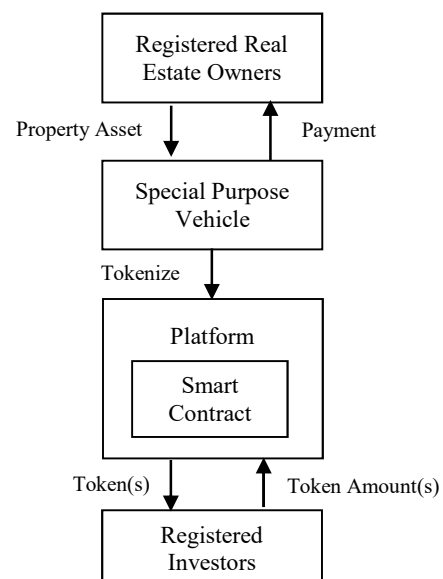


Fig. 5. ERC-777 Tokenization of Real Estate mechanism flow. From Gupta et al. (2020). *Tokenization of Real Estate Using Blockchain Technology*.

Real estate owners and investors should go through regulatory KYC/AML document verification as a part of registration process on the platform. From there, both parties can interact with each other and perform transactions on the platform.

On the other hand, ERC-1155 unifies the roles of ERC-20 and ERC-721 by supporting both fungible and non-fungible tokens in a single contract. This reduces the need for multiple deployments and enables batch operations, leading to significant gas savings.

S. Joshi emphasize the applicability of ERC-1155 in tokenizing real estate assets, where fractional ownership and unique asset identifiers can coexist efficiently. Compared to ERC-721, which requires one contract per NFT, ERC-1155 improves scalability and simplifies asset management through shared logic and metadata referencing.

Similar to A. Gupta's paper, S. Joshi's discussion also implied a different approach, whereas they proposed a description of workflow mechanism using the ERC-1155 as shown in "Fig. 6" in the form of a flowchart diagram.

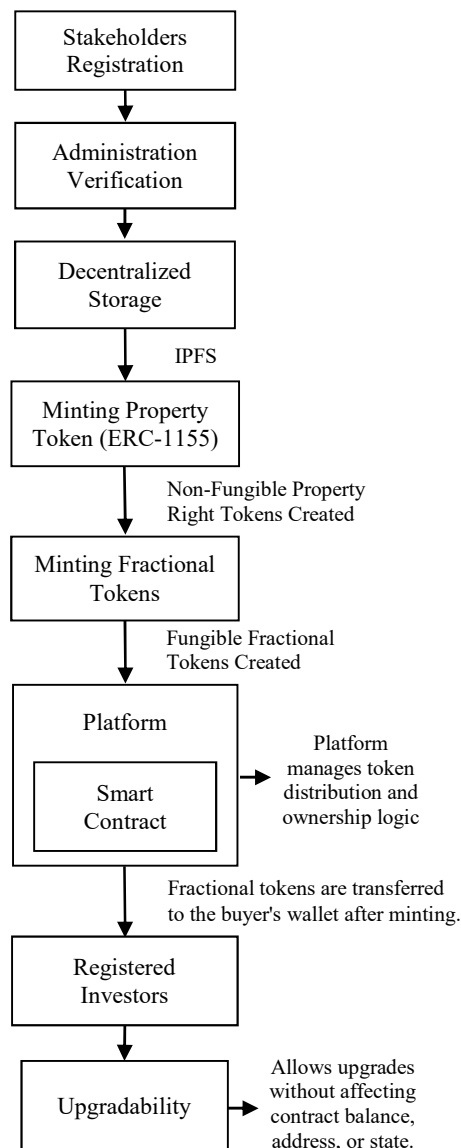


Fig. 6. ERC-1155 proposed interaction and processes.
Adapted from Joshi, S. & Choudhury, A. (2022). *Tokenization of Real Estate Assets Using Blockchain*.

Combined, ERC-777 and ERC-1155 provide complementary solutions: ERC-777 enables compliance automation, while ERC-1155 supports scalable multi-asset management. This thesis contributes by integrating these features into a proposed framework that explicitly links newer standards to gaps in existing real estate tokenization mechanisms.

These insights form the basis for the concluding remarks in the next section, where this thesis consolidates its comparative analysis and proposed framework into a clear contribution for both researchers and practitioners.

V. CONCLUSION

This study examined real estate tokenization mechanisms and explained why ERC-20, ERC-721, and ERC-1400 remain the dominant standards due to their maturity, simplicity, and regulatory alignment. However, these standards also reveal clear gaps in compliance automation, flexibility, and multi-asset management. Beyond its systematic review, the novelty of this thesis lies in its proposed framework that positions ERC-777 and ERC-1155 as solutions to these limitations. Specifically, ERC-777 addresses compliance and investor protection through automated token-level checks, while ERC-1155 supports scalable management of diverse real estate assets under a unified contract.

While prior works (e.g., Gupta 2020; Joshi 2022) demonstrated the feasibility of applying these standards to real estate, this thesis advances the discussion by explicitly linking them to the shortcomings of earlier implementations. This contribution provides a forward-looking roadmap for researchers, developers, and policymakers to consider ERC-777 and ERC-1155 in designing the next generation of tokenized real estate platforms.

Ultimately, these insights reflect the original hopes of investors: to make real estate more accessible, liquid, and transparent through blockchain innovation.

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